Why Hierarchical Attention Works: A Declared-Package Fiber-Diameter Diagnostic

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Abstract

Once a viewpoint (declared package) V is fixed, distinguishability—and therefore the logical validity of treating an endpoint representation as "sufficient" on an admissible class $\mathcal{C}$—is a V-conditioned statement.We give a fixed-slice (within-endpoint) diagnostic for hierarchical and multi-scale attention via fiber diameters and global gaps. A declared package $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ locks an admissible history class $\mathcal{C} \subseteq \mathcal{H}$, a query family $\mathcal{Q}$ (instantiated here as the singleton next-step prediction query), the reporting threshold ε, and an interface module $\mathcal{I} = (\Pi, D_{\mathrm{rep}}, U, D_{\mathrm{aux}})$ declaring the protocol-visible predictive interface, the reportable discrepancy readout (TV by default), admissible reconstructions, and any licensed auxiliary readouts.Given an endpoint map $\mathrm{End} : \mathcal{H} \to \mathcal{E}$, histories are partitioned into fibers $\mathrm{Fib}(e) = \mathrm{End}^{-1}(e)$. For a readout D (TV by default), we define fiber diameters $\mathrm{Diam}_D(e; \mathcal{C})$ and the global gap $\mathrm{Gap}_D(\mathrm{End}; \mathcal{C}) = \sup_{e \in \mathrm{End}(\mathcal{C})} \mathrm{Diam}_D(e; \mathcal{C})$, quantifying worst-case within-fiber predictive non-constancy. We report ε-sufficiency under V by the convention $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}; \mathcal{C}) < \varepsilon$, rather than as an identity claim.For refining endpoint families $\{\mathrm{End}_K\}$, fibers become thinner and the global gap is non-increasing. When KL is licensed under V, Pinsker-type bounds translate KL-scale reasoning into TV-scale conclusions. We relate $\mathrm{Gap}_{\mathrm{TV}}$ to a fixed-slice minimax reconstruction residual $\mathrm{Res}_{\mathrm{TV}}(\mathrm{End}; \mathcal{C}, U_{\mathcal{E}})$ and give packing-based lower bounds showing when coarse endpoints cannot be ε-sufficient on $\mathcal{C}$. Optional lightweight Python illustrations provide declared proxy visualizations on controlled processes (no model training required).

1 Introduction

Hierarchical and multi-scale mechanisms—including local-window attention [2], dilated/log-sparse patterns [3, 8, 18], hierarchical token summaries [7, 17], and compressed memory tiers [6, 10]—are widely used to extend sequence models [13] to long contexts under finite compute. While empirical results are abundant, the structural question "why does this class of mechanisms work?" is often addressed informally, typically by appeals to increased receptive fields or additional global context. A common implicit claim is that such endpoints retain all information that remains distinguishable to the declared predictive protocol on the working context window. We make this claim explicit and turn it into a package-conditional, testable criterion.

This paper isolates a minimal diagnostic core under a declared viewpoint. We fix a *declared package* $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$, an audit bundle locking an admissible history class $\mathcal{C} \subseteq \mathcal{H}$, a query family $\mathcal{Q}$, the reporting threshold ε, and an interface module $\mathcal{I} = (\Pi, D_{\mathrm{rep}}, U, D_{\mathrm{aux}})$ declaring the protocol-visible predictive interface, the reportable discrepancy readout (TV by default), admissible reconstructions, and any auxiliary readouts (subject to licensing under V).All statements are read as V-conditioned on $\mathcal{C}$; when an auxiliary readout is not licensed under V, the corresponding

gap/diameter is treated as undefined rather than approximated.Under this lock, the logical validity of treating an endpoint as "sufficient" is a statement about reportable discrepancy on $\mathcal{C}$ relative to the threshold ε, rather than a universal identification claim. Throughout the main text we adopt the standing single-query convention of Definition 1 and suppress explicit query arguments in the notation.

Given an endpoint map $\mathrm{End} : \mathcal{H} \to \mathcal{E}$, each endpoint value $e \in \mathcal{E}$ induces a *fiber* $\mathrm{Fib}(e) = \mathrm{End}^{-1}(e)$, i.e., an equivalence class of histories indistinguishable at the endpoint level. This paper is fixed-slice: all comparisons are within a fiber at a single endpoint value, not along a path in scale/hierarchy. We quantify within-endpoint residual ambiguity using fiber diameters $\mathrm{Diam}_D(e; \mathcal{C})$ and the global gap $\mathrm{Gap}_D(\mathrm{End}; \mathcal{C})$, i.e., the worst-case discrepancy between predictive outputs among histories in $\mathcal{C}$ that share the same endpoint. In this framework, hierarchical/multi-scale mechanisms are interpreted as *endpoint families* $\{\mathrm{End}_K\}$; when the family is refining on $\mathcal{C}$, it progressively refines the coarse description, thereby thinning fibers and (monotonically) reducing within-fiber predictive non-constancy as measured by the global gap. In this sense, the slogan "hierarchical attention works" is read here as the existence of refinements driving $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}_K; \mathcal{C})$ below ε under the declared package V.

Two principles make the framework concise. First, a refinement monotonicity statement shows that if End_{K+1} refines End_K, then the TV gap is non-increasing, and (when the KL readout is licensed under V) the KL gap is also non-increasing. Second, a KL-to-TV bridge (Pinsker-type) allows one to reason using KL when convenient and report conclusions in TV, which is bounded and directly interpretable as event-level probability discrepancy. We also state a minimal necessity obstruction: if the endpoint space is too small relative to the intrinsic diversity of predictive distributions, a nontrivial fiber gap must persist (via packing/entropy arguments; see, e.g., [9, 14]).

Relation to prior work (non-required). This paper is self-contained and studies within-endpoint (fixed-slice) fiber diagnostics under a declared package V. A brief notation-alignment and scope-boundary summary relative to [20, 19] is collected in Appendix A.

The scope of this paper is diagnostic rather than prescriptive. The goal is to provide a package-conditional language that (i) formalizes ε-sufficient endpoint representations as those for which the TV gap falls below ε under V (a reporting convention), and (ii) clarifies when refinements can or cannot eliminate within-fiber predictive ambiguity on $\mathcal{C}$.To keep the exposition self-contained, we include optional lightweight computational illustrations on controlled processes where $p(\cdot \mid h)$ is defined explicitly and no model training is required.

Contributions.

- A declared-package diagnostic language: endpoint maps, fibers, and fiber gaps (TV/KL fiber diameters).
- A refinement monotonicity principle: endpoint refinement implies non-increasing global gaps.
- A KL-to-TV bridge enabling TV-scale conclusions with minimal overhead.
- A simple capacity obstruction via packing: lower bounds showing when coarse endpoints cannot be ε-sufficient under fixed V.
- Optional lightweight computational illustrations (no model training required) demonstrating aligned refinement versus blind-spot plateaus.

2 Declared-Package Diagnostic Setup

2.1 Declared package V

Histories and output spaces. Fix a token alphabet $\mathcal{X}$. Let $\mathcal{H} := \bigcup_{T\geq 0} \mathcal{X}^T$ denote the set of finite histories, equipped with the length map $|h| = T$ for $h = (x_0, \ldots, x_{T-1})$. For $T \geq 0$ we write $\mathcal{X}^{\leq T} := \bigcup_{t=0}^{T} \mathcal{X}^t$. The predictive interface returns distributions over an output space $\mathcal{Y}$ (for next-token prediction one typically takes $\mathcal{Y} = \mathcal{X}$, but the diagnostic treats $\mathcal{Y}$ abstractly).

Definition 1 (Declared package)**.** A declared package is a tuple

$$V := (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C}),$$

where $\mathcal{C} \subseteq \mathcal{H}$ is an admissible history class (domain gate), $\mathcal{Q}$ is an admissible query family, and $\varepsilon > 0$ is a reporting threshold specifying the convention "$\approx_\varepsilon 0$" for reportable discrepancies.$\mathcal{I}$ is an interface module collecting the protocol-visible objects required by the diagnostic. In this paper, we take

$$\mathcal{I} := (\Pi, D_{\text{rep}}, U, D_{\text{aux}}),$$

where $\mathcal{P}(\mathcal{Y})$ denotes the output distribution space over $\mathcal{Y}$, and $\Pi : \mathcal{H} \times \mathcal{Q} \to \mathcal{P}(\mathcal{Y})$ denotes the protocol-visible predictive interface. We write its output as $p(\cdot \mid h, q) := \Pi(h, q) \in \mathcal{P}(\mathcal{Y})$. **Standing convention (single query).** Fix a distinguished query $q_\star$ and set $\mathcal{Q} = \{q_\star\}$. We abbreviate $\Pi(h) := \Pi(h, q_\star)$ and $p(\cdot \mid h) := \Pi(h)$; all gaps/diameters and licensing statements are evaluated on these induced predictions. The declared *reportable* discrepancy functional on $\mathcal{P}(\mathcal{Y})$ is D_{rep} (TV by default), and D_{aux} denotes an optional collection of declared auxiliary readouts on $\mathcal{P}(\mathcal{Y})$ (e.g. relative entropy, when licensed under V). Finally, U is a declared admissible lifting family for endpoint-to-distribution reconstruction. Formally, $U = \{U_E\}_E$ is an endpoint-space–indexed family such that for each endpoint value space E one has $U_E \subseteq \{u : E \to \mathcal{P}(\mathcal{Y})\}$. Once an endpoint map $\text{End} : \mathcal{H} \to \mathcal{E}$ is fixed, all fixed-slice reconstruction residuals take the infimum over $u \in U_{\mathcal{E}}$. Unless otherwise stated, we use the default $U_E = U_E^{\text{all}} := \{u : E \to \mathcal{P}(\mathcal{Y})\}$. An endpoint map $\text{End} : \mathcal{H} \to \mathcal{E}$ is a representation choice evaluated under the fixed package V; all fiber objects (Fib, Diam, Gap, and the fixed-slice residual) are indexed by End and reported under V.Notation. When convenient we abbreviate the reportable clauses of V by $(\mathcal{Q}, D_{\text{rep}}, \varepsilon)$, but no new object is introduced.

Package semantics (domain gate and reporting threshold). All optimization operators (sup/inf) in the diagnostic are taken over $\mathcal{C}$ unless explicitly stated otherwise. All diameters/gaps and residuals use the induced predictions $p(\cdot \mid h) = \Pi(h)$ declared in $\mathcal{I}$; once V is fixed we suppress this $\mathcal{I}$-dependence in notation. Under V we adopt the reporting convention

$$\text{Gap}_{\text{TV}}(\text{End}; \mathcal{C}) \approx_\varepsilon 0 \quad :\Leftrightarrow \quad \text{Gap}_{\text{TV}}(\text{End}; \mathcal{C}) < \varepsilon,$$

with TV understood as the reportable readout D_{rep}. This is the true-object reporting criterion $\text{Gap}_{\text{TV}}(\text{End}; \mathcal{C}) < \varepsilon$ under V.Here "$\approx_\varepsilon 0$" is a resolution-aware reporting convention (*reportable as 0 under* V), not an ontological identity.Accordingly, we use the thresholded notion "ε-sufficient under V" rather than binary labels.

Licensing convention. Fix an endpoint map $\text{End} : \mathcal{H} \to \mathcal{E}$. A declared discrepancy readout D on $\mathcal{P}(\mathcal{Y})$ is *licensed under* V *for* End *on* $\mathcal{C}$ if $D\big(p(\cdot \mid h_1), p(\cdot \mid h_2)\big)$ is well-defined and finite for all history pairs $(h_1, h_2) \in \mathcal{C} \times \mathcal{C}$ with $\text{End}(h_1) = \text{End}(h_2)$. In this paper, TV is always licensed; KL is licensed only under the mutual absolute-continuity condition in Remark 4. We use the total-variation distance $\text{TV}(p, q) := \sup_A |p(A) - q(A)|$, which satisfies $\text{TV} \in [0, 1]$; for discrete $\mathcal{Y}$, $\text{TV}(p, q) = \frac{1}{2}\|p - q\|_1$.

Default instantiation (used in illustrations). In the optional computational illustrations, $\mathcal{Q}$ contains only the next-step prediction query, the protocol-level reportable metric is fixed to $D_{\text{rep}} = \text{TV}$, and $\mathcal{C}$ is the fixed-length history class used for sampling (e.g., $\mathcal{C} = \{0,1\}^T$ with $T = 256$). Relative-entropy quantities appear only as auxiliary readouts in D_{aux} (e.g., for Pinsker-type reasoning), and are not used unless the corresponding readout is licensed (see Remark 4).

2.2 Predictive distributions

Let $\Pi : \mathcal{H} \times \mathcal{Q} \to \mathcal{P}(\mathcal{Y})$ denote the protocol-visible predictive interface. Under the standing single-query convention in Definition 1, we write $p(\cdot \mid h) := \Pi(h) \in \mathcal{P}(\mathcal{Y})$. In sequence modeling, h can be taken as the full prefix (or any internal state summarizing the prefix), and $p(\cdot \mid h)$ as the next-symbol distribution. Under the standing single-query convention in Definition 1, we write $p(\cdot \mid h) := \Pi(h)$ for the next-step predictive distribution and suppress explicit query arguments throughout.

LLM interpretation. For an autoregressive language model with vocabulary $\mathcal{Y}$, $p(\cdot \mid h)$ can be instantiated as the next-token distribution $\text{softmax}(W\, s(h))$, where $s(h)$ denotes the model's internal representation of the prefix [13]. In this paper, we treat Π abstractly as a protocol-visible predictive interface and $p(\cdot \mid h)$ as its protocol-visible output; the framework does not assume access to model internals.

Key point. The diagnostic focuses on how coarse descriptions of history (endpoints) constrain $p(\cdot \mid h)$. Once V is fixed, fiber gaps quantify the extent to which the endpoint does not determine the predictive distribution, thereby providing a structural lens for analyzing hierarchical and multi-scale mechanisms.

3 Endpoints, Fibers, and Fiber Diameters

Definition 2 (Endpoint map and fiber)**.** An endpoint map is a function $\text{End} : \mathcal{H} \to \mathcal{E}$. For $e \in \mathcal{E}$, the fiber over e is

$$\text{Fib}(e) := \text{End}^{-1}(e) = \{h \in \mathcal{H} : \text{End}(h) = e\}.$$

Effective endpoint image under $\mathcal{C}$. For a fixed domain gate $\mathcal{C} \subseteq \mathcal{H}$, only endpoint values attained on $\mathcal{C}$ are relevant. We therefore write

$$\mathcal{E}_{\mathcal{C}} := \text{End}(\mathcal{C}) = \{\text{End}(h) : h \in \mathcal{C}\} \subseteq \mathcal{E}.$$

All suprema over endpoint values in the diagnostic may be taken over $\mathcal{E}_{\mathcal{C}}$ without loss, thereby avoiding empty fibers.

Definition 3 (Fiber diameters and global gaps (domain-gated))**.** Write $D_{\text{KL}}^{\max}(p,q) := \max\{\text{KL}(p \,\|\, q)\,,\, \text{KL}(q \,\|\, p)\}$. Fix an endpoint map $\text{End} : \mathcal{H} \to \mathcal{E}$, an admissible class $\mathcal{C} \subseteq \mathcal{H}$, and a discrepancy readout $D : \mathcal{P}(\mathcal{Y}) \times \mathcal{P}(\mathcal{Y}) \to [0,\infty]$. For each endpoint $e \in \mathcal{E}_{\mathcal{C}}$, define the fiber diameter

$$\text{Diam}_D(e;\mathcal{C}) := \sup_{h_1,h_2 \in \text{Fib}(e)\cap\mathcal{C}} D\big(p(\cdot \mid h_1), p(\cdot \mid h_2)\big).$$

Define the corresponding global gap

$$\text{Gap}_D(\text{End};\mathcal{C}) := \sup_{\substack{h_1,h_2 \in \mathcal{C}: \\ \text{End}(h_1)=\text{End}(h_2)}} D\big(p(\cdot \mid h_1), p(\cdot \mid h_2)\big).$$

Remark 1 (Instances used in this paper). We treat total variation as the default reportable readout $D_{\mathrm{rep}} = \mathrm{TV}$ and write $\mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C}) := \mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C})$, $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) := \mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$. For relative entropy we use the symmetric max readout $D_{\mathrm{KL}}^{\max}$ and write $\mathrm{Diam}_{\mathrm{KL}}(e;\mathcal{C}) := \mathrm{Diam}_{D_{\mathrm{KL}}^{\max}}(e;\mathcal{C})$, $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C}) := \mathrm{Gap}_{D_{\mathrm{KL}}^{\max}}(\mathrm{End};\mathcal{C})$.

Remark 2 (Totality convention for singleton fibers). In the suprema defining diameters and gaps we allow $h_1 = h_2$. In particular, fiber diameters are 0 on singleton fibers, and the global gaps are well-defined on $\mathcal{E}_{\mathcal{C}}$.

Lemma 1 (Fiber decomposition)**.** *For any endpoint map* $\mathrm{End} : \mathcal{H} \to \mathcal{E}$*, any* $\mathcal{C} \subseteq \mathcal{H}$*, and any readout* D*,*

$$\mathrm{Gap}_D(\mathrm{End};\mathcal{C}) = \sup_{e\in\mathcal{E}_{\mathcal{C}}} \mathrm{Diam}_D(e;\mathcal{C}).$$

Proof. For any $h_1, h_2 \in \mathcal{C}$ with $\mathrm{End}(h_1) = \mathrm{End}(h_2) = e$, the discrepancy contributes to $\mathrm{Diam}_D(e;\mathcal{C})$. Taking a supremum over all such pairs and then over e yields $\mathrm{Gap}_D(\mathrm{End};\mathcal{C}) \le \sup_{e\in\mathcal{E}_{\mathcal{C}}} \mathrm{Diam}_D(e;\mathcal{C})$. Conversely, each $\mathrm{Diam}_D(e;\mathcal{C})$ is a supremum over a subset of the pairs used in $\mathrm{Gap}_D(\mathrm{End};\mathcal{C})$, so $\mathrm{Diam}_D(e;\mathcal{C}) \le \mathrm{Gap}_D(\mathrm{End};\mathcal{C})$ for all $e \in \mathcal{E}_{\mathcal{C}}$, hence $\sup_{e\in\mathcal{E}_{\mathcal{C}}} \mathrm{Diam}_D(e;\mathcal{C}) \le \mathrm{Gap}_D(\mathrm{End};\mathcal{C})$. □

Remark 3 (TV as the final presentation metric). TV is bounded in $[0, 1]$ and directly controls event-level probability differences, making it convenient for reporting and figures [4, 12]. Relative entropy is used when chain rules or information monotonicity are needed [4, 5].

Remark 4 (Licensing gate for relative entropy and auxiliary readouts). Here TV is treated as the reportable readout D_{rep}; relative entropy is an auxiliary readout in D_{aux} and may be non-finite unless an absolute-continuity condition holds. Accordingly, the KL-based readouts are *licensed under* V on $\mathcal{C}$ only when, for all endpoints $e \in \mathcal{E}_{\mathcal{C}}$ and all $h_1, h_2 \in \mathrm{Fib}(e) \cap \mathcal{C}$, both directed values

$$\mathrm{KL}(p(\cdot \mid h_1) \,\|\, p(\cdot \mid h_2)) \quad \text{and} \quad \mathrm{KL}(p(\cdot \mid h_2) \,\|\, p(\cdot \mid h_1))$$

are finite (equivalently, the pair is mutually absolutely continuous on the support relevant under V). Under this licensing condition, the KL-based diameters/gaps in this paper use the symmetric max readout $D_{\mathrm{KL}}^{\max}$ as defined in Definition 3, so that, e.g., $\mathrm{Diam}_{\mathrm{KL}}(e;\mathcal{C}) = \sup_{h_1,h_2\in\mathrm{Fib}(e)\cap\mathcal{C}} D_{\mathrm{KL}}^{\max}(p(\cdot \mid h_1), p(\cdot \mid h_2))$, and similarly for $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})$.

If this condition does not hold, then KL-based diameters/gaps are *not licensed* (undefined) under V, and TV remains the reportable metric.

In the optional Monte Carlo illustrations, we therefore declare a finite auxiliary readout $\mathrm{KL}_{\mathrm{clip(EPS)}}$ obtained by EPS-clipping (see Section 6.3), and we report only the corresponding proxy $\widehat{\mathrm{Gap}}_{\mathrm{KL,clip(EPS)},N}(\mathrm{End};\mathcal{C})$.

Here EPS is a numerical regularization parameter and is unrelated to the reporting threshold ε declared in V.

Important: $\mathrm{KL}_{\mathrm{clip(EPS)}}$ is a separate declared numerical readout and is never treated as licensed KL in any theorem or TV bridge.

4 Structural Results

4.1 Endpoint Refinement and Monotonicity

Definition 4 (Endpoint refinement on $\mathcal{C}$)**.** An endpoint map $\mathrm{End}_2 : \mathcal{H} \to \mathcal{E}_2$ refines $\mathrm{End}_1 : \mathcal{H} \to \mathcal{E}_1$ on $\mathcal{C}$ if

$$\mathrm{End}_2(h) = \mathrm{End}_2(h') \;\Rightarrow\; \mathrm{End}_1(h) = \mathrm{End}_1(h') \quad \text{for all } h, h' \in \mathcal{C}.$$

Proposition 1 (Refinement reduces gaps (domain-gated))**.** *If* End_2 *refines* End_1 *on* $\mathcal{C}$*, then for any discrepancy readout* D *such that both* $\mathrm{Gap}_D(\mathrm{End}_1;\mathcal{C})$ *and* $\mathrm{Gap}_D(\mathrm{End}_2;\mathcal{C})$ *are licensed under* V *(for* End_1 *and* End_2 *respectively),*

$$\mathrm{Gap}_D(\mathrm{End}_2;\mathcal{C}) \le \mathrm{Gap}_D(\mathrm{End}_1;\mathcal{C}).$$

Proof. The set $\{(h_1,h_2) \in \mathcal{C}\times\mathcal{C} : \mathrm{End}_2(h_1) = \mathrm{End}_2(h_2)\}$ is a subset of $\{(h_1,h_2) \in \mathcal{C}\times\mathcal{C} : \mathrm{End}_1(h_1) = \mathrm{End}_1(h_2)\}$. Taking a supremum over a smaller set cannot increase the value. □

Remark 5 (Licensing inheritance under refinement)*.* For $D = \mathrm{TV}$, licensing is automatic. For KL-based readouts, if $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End}_1;\mathcal{C})$ is licensed under V, then $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End}_2;\mathcal{C})$ is also licensed whenever End_2 refines End_1 on $\mathcal{C}$, since the compared history-pair set shrinks under refinement.

4.2 KL-to-TV Bridge

Proposition 2 (Pinsker bridge)**.** *Recall the symmetric max-KL readout* $D_{\mathrm{KL}}^{\max}$ *from Definition 3. For any distributions* $p, q \in \mathcal{P}(\mathcal{Y})$*,*

$$\mathrm{TV}(p,q) \le \sqrt{\tfrac{1}{2}\,\mathrm{KL}(p \,\|\, q)} \;\le\; \sqrt{\tfrac{1}{2}\,D_{\mathrm{KL}}^{\max}(p,q)},$$

with the convention $\sqrt{\infty} = \infty$*. This inequality is standard; see, e.g., [4, 12].*

Consequently, if $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})$ *is licensed under* V *for* End *on* $\mathcal{C}$ *and finite (i.e., induced by* $D_{\mathrm{KL}}^{\max}$ *under the licensing gate in Remark 4), then*

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) \le \sqrt{\tfrac{1}{2}\,\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})}.$$

Scope of the bridge. *This proposition is invoked only when* $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})$ *is licensed under* V*. Declared surrogate readouts (e.g. clipping-based proxies used in illustrations) are not treated as licensed KL and do not yield TV bounds for the original distributions.*

4.3 Interpretation for Hierarchical / Multi-Scale Mechanisms

We interpret hierarchical and multi-scale mechanisms as *endpoint families*; when such a family is refining on $\mathcal{C}$, it progressively refines a coarse description of history. Fix a declared package $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ that locks the protocol-visible predictive interface Π, the reportable readout D_{rep}, the admissible lifting family U, and any auxiliary readouts D_{aux}.For a given endpoint map $\mathrm{End} : \mathcal{H} \to \mathcal{E}$, each endpoint value $e \in \mathcal{E}$ induces a fiber $\mathrm{Fib}(e) = \mathrm{End}^{-1}(e)$. The quantity $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$ measures the worst-case *within-fiber predictive non-constancy* on $\mathcal{C}$: how different the predictive distributions can remain even when the endpoint agrees.

Diagnostic criterion (package-specific). Under V, the statement

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) < \varepsilon$$

formalizes that the endpoint is ε-sufficient under the declared package V (TV-scale, on $\mathcal{C}$).This is a thresholded statement rather than an absolute one.

Minimax reconstruction residual (notation $\mathrm{Res}_{\mathrm{TV}}$). In addition to $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$, it is useful to quantify how well one can reconstruct a representative predictive distribution from the endpoint alone, under an admissible reconstruction family U. This is a fixed-slice (within-endpoint) residual local to the present diagnostic and should not be confused with the endpoint–path bare symbols Δ_I and Γ_{path} reserved in Section A.

Definition 5 (Fixed-slice minimax residual ($\mathrm{Res}_{\mathrm{TV}}$ instance))**.** Fix an endpoint map $\mathrm{End}:\mathcal{H}\to\mathcal{E}$. Let $U_{\mathcal{E}}\subseteq\{u:\mathcal{E}\to\mathcal{P}(\mathcal{Y})\}$ be the declared admissible reconstruction class over $\mathcal{E}$ from the lifting family U in $\mathcal{I}$ (default $U_{\mathcal{E}}=U_{\mathcal{E}}^{\mathrm{all}}$). Define the reportable TV residual

$$\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}}) := \inf_{u\in U_{\mathcal{E}}}\ \sup_{h\in\mathcal{C}}\ \mathrm{TV}(p(\cdot\mid h),u(\mathrm{End}(h)))\,.$$

Lemma 2 (Residual versus gap)**.** *Fix an endpoint map* $\mathrm{End}:\mathcal{H}\to\mathcal{E}$*. For TV readout:*

1. *For any nonempty admissible reconstruction class $U_{\mathcal{E}}$, one always has*

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})\ \le\ 2\,\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}}).$$

2. *If $U_{\mathcal{E}}$ contains all fiber-selector reconstructions (i.e., maps u for which there exists a choice of representatives $h_e\in\mathrm{Fib}(e)\cap\mathcal{C}$ for each $e\in\mathcal{E}_{\mathcal{C}}$ such that $u(e)=p(\cdot\mid h_e)$), then*

$$\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}})\ \le\ \mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}).$$

In the default case $U_{\mathcal{E}}=U_{\mathcal{E}}^{\mathrm{all}}$, the two bounds combine to

$$\tfrac{1}{2}\,\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})\ \le\ \mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}})\ \le\ \mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}).$$

Under the selector condition in (2), selecting for each $e\in\mathcal{E}_{\mathcal{C}}$ an arbitrary representative $h_e\in\mathrm{Fib}(e)\cap\mathcal{C}$ and setting $u(e):=p(\cdot\mid h_e)$ (extended arbitrarily to all of $\mathcal{E}$) yields

$$\sup_{h\in\mathrm{Fib}(e)\cap\mathcal{C}}\ \mathrm{TV}(p(\cdot\mid h),u(e))\ \le\ \mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C}).$$

Proof. (*Upper bound.*) Assume $U_{\mathcal{E}}$ contains fiber-selector reconstructions (in particular, this holds under the default $U_{\mathcal{E}}=U_{\mathcal{E}}^{\mathrm{all}}$). Fix any representatives $h_e\in\mathrm{Fib}(e)\cap\mathcal{C}$ for each $e\in\mathcal{E}_{\mathcal{C}}$ and define $u(e):=p(\cdot\mid h_e)$ on $\mathcal{E}_{\mathcal{C}}$, extending u arbitrarily to all of $\mathcal{E}$. By assumption, this u lies in $U_{\mathcal{E}}$. For any $h\in\mathrm{Fib}(e)\cap\mathcal{C}$, by definition of $\mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C})$ we have $\mathrm{TV}(p(\cdot\mid h),u(e))\le\mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C})$. Taking a supremum over $h\in\mathcal{C}$ and using Lemma 1 gives $\sup_{h\in\mathcal{C}}\mathrm{TV}(p(\cdot\mid h),u(\mathrm{End}(h)))\le\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$. Taking an infimum over $u\in U_{\mathcal{E}}$ yields $\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}})\le\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$.

(*Lower bound.*) Fix any $u\in U_{\mathcal{E}}$. For each $e\in\mathcal{E}_{\mathcal{C}}$, define the fiber radius

$$R_u(e) := \sup_{h\in\mathrm{Fib}(e)\cap\mathcal{C}}\ \mathrm{TV}(p(\cdot\mid h),u(e))\,.$$

For any $h_1,h_2\in\mathrm{Fib}(e)\cap\mathcal{C}$, the triangle inequality gives

$$\mathrm{TV}(p(\cdot\mid h_1),p(\cdot\mid h_2))\le\mathrm{TV}(p(\cdot\mid h_1),u(e))+\mathrm{TV}(p(\cdot\mid h_2),u(e))\le 2\,R_u(e).$$

Taking a supremum over $h_1,h_2\in\mathrm{Fib}(e)\cap\mathcal{C}$ yields $\mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C})\le 2\,R_u(e)$, hence

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})=\sup_{e\in\mathcal{E}_{\mathcal{C}}}\mathrm{Diam}_{\mathrm{TV}}(e;\mathcal{C})\le 2\sup_{e\in\mathcal{E}_{\mathcal{C}}}R_u(e)\le 2\sup_{h\in\mathcal{C}}\mathrm{TV}(p(\cdot\mid h),u(\mathrm{End}(h)))\,.$$

Taking an infimum over $u\in U_{\mathcal{E}}$ yields $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})\le 2\,\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C},U_{\mathcal{E}})$. □

Hierarchical refinement as fiber thinning. A hierarchical mechanism augments a baseline endpoint End_0 (e.g., a local window) with additional multi-scale summaries, memory tiers, or sparse long-range anchors, yielding a family $\{\mathrm{End}_K\}_{K\geq 0}$. When this augmentation is implemented by concatenating additional deterministic coordinates to the endpoint, End_K refines End_{K-1} by construction. We allow the endpoint codomain to depend on K: $\mathrm{End}_K : \mathcal{H} \to \mathcal{E}_K$. For each fixed K, define the domain-gated endpoint image

$$\mathcal{E}_{\mathcal{C}}^{(K)} := \mathrm{End}_K(\mathcal{C}) \subseteq \mathcal{E}_K.$$

All fiber quantities for End_K may take suprema over $e \in \mathcal{E}_{\mathcal{C}}^{(K)}$, and the fixed-slice residual uses the corresponding reconstruction class $U_{\mathcal{E}_K}$. By construction in the deterministic-concatenation setting, End_{K+1} refines End_K, so Proposition 1 implies

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}_{K+1};\mathcal{C}) \leq \mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}_K;\mathcal{C}).$$

Moreover, whenever the KL readout is licensed under V, one also has

$$\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End}_{K+1};\mathcal{C}) \leq \mathrm{Gap}_{\mathrm{KL}}(\mathrm{End}_K;\mathcal{C}).$$

Thus, adding scales can be read as a structural operation that shrinks fibers and (weakly) reduces within-fiber predictive variation on $\mathcal{C}$.

Why TV for presentation and KL for reasoning. TV is bounded in $[0, 1]$ and directly controls event-level probability differences, making it suitable for reporting and figures. KL aligns with log-loss objectives and supports information-theoretic manipulations [4, 5]; Proposition 2 provides a one-line bridge back to TV-scale conclusions.

4.4 A simple necessity principle (capacity lower bound)

Refinement guarantees monotone *improvement*, but does not guarantee that gaps can be driven below a target ε.A minimal obstruction is endpoint capacity.

Definition 6 (TV packing number)**.** Let $\mathcal{P}_{\mathcal{C}} := \{p(\cdot \mid h) : h \in \mathcal{C}\} \subseteq \mathcal{P}(\mathcal{Y})$. For $\delta > 0$, define $\mathsf{Pack}_{\mathrm{TV}}(\delta;\mathcal{P}_{\mathcal{C}})$ as the maximal cardinality of a subset $\{p_i\} \subseteq \mathcal{P}_{\mathcal{C}}$ such that $\mathrm{TV}(p_i, p_j) > \delta$ for all $i \neq j$.

Packing/entropy notions are classical; see, e.g., [9, 14].

Proposition 3 (Pigeonhole lower bound)**.** *Fix an endpoint map* $\mathrm{End} : \mathcal{H} \to \mathcal{E}$ *and assume* $\mathcal{E}_{\mathcal{C}} := \mathrm{End}(\mathcal{C})$ *is finite. Fix* $\delta > 0$*. If* $|\mathcal{E}_{\mathcal{C}}| < \mathsf{Pack}_{\mathrm{TV}}(\delta;\mathcal{P}_{\mathcal{C}})$*, then*

$$\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) \geq \delta.$$

Proof. Choose $\{p_i\}_{i=1}^m \subseteq \mathcal{P}_{\mathcal{C}}$ with $m = \mathsf{Pack}_{\mathrm{TV}}(\delta;\mathcal{P}_{\mathcal{C}})$ and pairwise TV separation $> \delta$. Pick histories $h_i \in \mathcal{C}$ such that $p(\cdot \mid h_i) = p_i$. Since $|\mathcal{E}_{\mathcal{C}}| < m$, two distinct histories h_i, h_j must share the same endpoint, i.e., $\mathrm{End}(h_i) = \mathrm{End}(h_j)$, yet $\mathrm{TV}(p(\cdot \mid h_i), p(\cdot \mid h_j)) > \delta$, implying $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) \geq \delta$. □

Remark 6 (Finiteness in the default instantiation). If $\mathcal{C}$ is finite (e.g., $\mathcal{C} = \{0, 1\}^T$ in the illustrations), then $\mathcal{E}_{\mathcal{C}} = \mathrm{End}(\mathcal{C})$ is automatically finite, so the finiteness assumption in Proposition 3 holds in the computational examples. If $\mathcal{E}_{\mathcal{C}}$ is infinite, Proposition 3 does not apply directly and must be replaced by a metric-entropy (effective-cardinality) obstruction.

5 Endpoint Instantiations for Hierarchical Attention

5.1 Single-Scale Endpoints (Local/Sliding Window)

A single-scale local-window endpoint retains only a bounded local view of the history. For a sequence-valued history $h = (x_0, \ldots, x_{T-1})$ over an alphabet $\mathcal{X}$, indexed 0-based so that x_{T-d} denotes the token d steps before the end (matching Python-style negative indexing), a canonical local endpoint is

$$\mathrm{End}_w^{\mathrm{loc}}(h) := (x_{T-w}, \ldots, x_{T-1}),$$

with the natural truncation convention when $T < w$. This endpoint class abstracts sliding-window attention [2]: states that agree on the last w tokens are placed in the same fiber.

5.2 Hierarchical Endpoints (Chunk Summaries / Multi-Resolution Tokens)

A hierarchical endpoint augments a baseline local view with multi-scale summaries. Let $\pi_s : \mathcal{X}^{\leq T} \to \mathcal{S}_s$ be a summary operator at scale s (e.g., hierarchical pooling/summarization [7, 17] or compressed memory tiers [10]). Define a family

$$\mathrm{End}_{w,K}^{\mathrm{hier}}(h) := \Big(\mathrm{End}_w^{\mathrm{loc}}(h),\ \pi_{s_1}(h),\ \pi_{s_2}(h),\ \ldots,\ \pi_{s_K}(h)\Big), \quad s_1 < s_2 < \cdots < s_K.$$

By construction, $\mathrm{End}_{w,K}^{\mathrm{hier}}$ refines $\mathrm{End}_{w,K-1}^{\mathrm{hier}}$ (equality of the K-tuple implies equality of its prefix), and therefore Gap_D is non-increasing in K by Proposition 1. This captures the structural content of hierarchical attention and memory compression [7, 17, 10]: additional scales thin fibers by adding coarse but discriminative coordinates.

5.3 Multi-Scale Sparse Endpoints (Dilated / Log-Stride Anchors)

Multi-scale sparse endpoints capture long-range information by sampling a small set of anchor positions. Introduce a sentinel symbol $\perp \notin \mathcal{X}$. For a history $h = (x_0, \ldots, x_{T-1})$, define the anchor-extraction map

$$\mathrm{Anch}(h, t) := \begin{cases} x_t, & 0 \leq t \leq T-1, \\ \perp, & \text{otherwise.} \end{cases}$$

Fix a base window width w and a growth factor $a > 1$. Let distances be $d_\ell := \lfloor w a^{\ell-1} \rfloor$ for $\ell = 1, \ldots, K$. A simple log-stride endpoint is

$$\mathrm{End}_{w,K}^{\mathrm{sparse}}(h) := \Big(\mathrm{End}_w^{\mathrm{loc}}(h),\ \mathrm{Anch}(h, T-d_1),\ \mathrm{Anch}(h, T-d_2), \ldots, \mathrm{Anch}(h, T-d_K)\Big),$$

so $\mathrm{End}_{w,K}^{\mathrm{sparse}}$ maps into $\mathcal{X}^{\leq w} \times (\mathcal{X} \cup \{\perp\})^K$. To mitigate phase blind spots, one may consider a small set of offsets $\{\sigma_r\}_{r=1}^R$ and include anchors $\mathrm{Anch}(h, T - d_\ell - \sigma_r)$. In this case, the endpoint should be interpreted as a deterministically ordered tuple (e.g., lexicographic in ℓ, r), mapping into $\mathcal{X}^{\leq w} \times (\mathcal{X} \cup \{\perp\})^{KR}$. This abstraction matches the structural role of dilated/log-sparse attention [3, 18, 8]: it refines a purely local endpoint using exponentially spaced observations.

6 Computational Illustration Protocol (No Training)

This section provides optional computational sanity checks that require no model training. The predictive distribution rule $h \mapsto p(\cdot \mid h)$ is defined explicitly by construction on controlled processes, and fiber gaps are estimated by bucketing sampled histories under a chosen endpoint map.

6.1 Controlled Long-Range Dependency Processes

We use binary next-token distributions for maximal transparency. Let $\mathcal{Y} = \{0, 1\}$ and let $p(\cdot \mid h)$ be a Bernoulli distribution with parameter $\theta(h) \in [0, 1]$. In this case,

$$\mathrm{TV}(p(\cdot \mid h_1), p(\cdot \mid h_2)) = |\theta(h_1) - \theta(h_2)|.$$

Process A (compressible long-range dependence). Let $h = (x_0, \ldots, x_{T-1}) \in \{0, 1\}^T$ and define a slow latent statistic

$$z(h) := \Big(\sum_{t=T-B}^{T-1} x_t \Big) \bmod 2,$$

where B is a large block size. Define

$$\theta_A(h) := \theta_0 + \alpha\, z(h),$$

and set $\theta(h) := \theta_A(h)$ for this process, so that the next-token distribution takes one of two well-separated values. Assume $\theta_0 \in [0, 1 - \alpha]$ so that $\theta_A(h) \in [0, 1]$ for all h. We take $\alpha \in (0, 1)$ as a scalar amplitude parameter of the toy process; it is unrelated to any gap notation.

This process is "compressible" in the sense that a single coarse statistic $z(h)$ controls the prediction.

Process B (blind-spot dependence). Fix a lag $L^\star$. Define a deterministic next-token rule

$$\theta_B(h) := \begin{cases} x_{T-L^\star}, & T \geq L^\star, \\ 0, & T < L^\star. \end{cases}$$

and set $\theta(h) := \theta_B(h)$ for this process. This process isolates a single critical coordinate; endpoints that do not observe $x_{T-L^\star}$ incur maximal within-fiber predictive variation.

6.2 Endpoint families used in the illustrations

We instantiate endpoints following Section 5: (i) local window $\mathrm{End}_w^{\mathrm{loc}}$; (ii) local plus exponential anchors $\mathrm{End}_{w,K}^{\mathrm{sparse}}$; (iii) local plus multi-scale block parities as a special case of $\mathrm{End}_{w,K}^{\mathrm{hier}}$ with $\pi_s(h) = \big(\sum_{t=T-s}^{T-1} x_t\big) \bmod 2$. These endpoints highlight a structural distinction: the TV proxy decreases only when the endpoint carries coordinates aligned with the predictive dependency.

6.3 Estimating fiber gaps by bucketing

Exact computation of $\mathrm{Gap}_D(\mathrm{End}; \mathcal{C})$ is generally intractable even for toy spaces, so we use a standard bucketing estimate.

Estimator notation. We denote this finite-sample proxy by $\widehat{\mathrm{Gap}}_{D,N}(\mathrm{End}; \mathcal{C})$ to emphasize its dependence on the sample size N (it is used only to display refinement trends, not as a package-licensed numerical report of $\mathrm{Gap}_D(\mathrm{End}; \mathcal{C})$). In particular, comparisons of $\widehat{\mathrm{Gap}}_{D,N}(\mathrm{End}; \mathcal{C})$ to ε are visual reference only and do not certify ε-sufficiency under V. Sample histories $h^{(1)}, \ldots, h^{(N)} \in \mathcal{C}$ and define the *observed endpoint image*

$$\mathcal{E}_N := \{\mathrm{End}(h^{(i)}) : i \in \{1, \ldots, N\}\} \subseteq \mathcal{E}_{\mathcal{C}}.$$

Group the samples by endpoint value $e \in \mathcal{E}_N$. For each bucket $e \in \mathcal{E}_N$, compute empirical extrema of the predictive parameter:

$$\theta_{\min}(e) := \min_{i:\mathrm{End}(h^{(i)})=e} \theta(h^{(i)}), \qquad \theta_{\max}(e) := \max_{i:\mathrm{End}(h^{(i)})=e} \theta(h^{(i)}).$$

For Bernoulli predictions, the maximum within-bucket TV is $\theta_{\max}(e) - \theta_{\min}(e)$. We then estimate the TV gap by the within-bucket range over observed buckets,

$$\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C}) := \max_{e\in\mathcal{E}_N} \big(\theta_{\max}(e) - \theta_{\min}(e)\big).$$

Smoothed relative-entropy readout (illustrations only). *EPS is a numerical regularization parameter and is unrelated to the reporting threshold ε declared in V.*Relative entropy can be infinite when probabilities hit 0 or 1. For numerical illustrations, we therefore fix a small EPS > 0 and define the clipping map

$$\mathrm{clip}_{\mathrm{EPS}}(p) := \min\{\max\{p, \mathrm{EPS}\}, 1 - \mathrm{EPS}\}.$$

For Bernoulli distributions, define the finite readout

$$\mathrm{KL}_{\mathrm{clip(EPS)}}\big(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(q)\big) := \mathrm{KL}(\mathrm{Bern}(\mathrm{clip}_{\mathrm{EPS}}(p)) \,\|\, \mathrm{Bern}(\mathrm{clip}_{\mathrm{EPS}}(q)))\,.$$

*This clipped readout is not a licensed KL divergence under V; it is an auxiliary numerical proxy for visualization and does not enter any formal claim or reporting criterion (which remains TV-based via D_{rep} and the convention $\approx_\varepsilon 0$).*We report the auxiliary estimate

$$\begin{aligned}\widehat{\mathrm{Gap}}_{\mathrm{KL,clip(EPS)},N}(\mathrm{End};\mathcal{C}) := \max_{e\in\mathcal{E}_N} \max\Big\{ & \\ & \mathrm{KL}_{\mathrm{clip(EPS)}}\Big(\mathrm{Bern}(\theta_{\min}(e)) \,\|\, \mathrm{Bern}(\theta_{\max}(e))\Big), \\ & \mathrm{KL}_{\mathrm{clip(EPS)}}\Big(\mathrm{Bern}(\theta_{\max}(e)) \,\|\, \mathrm{Bern}(\theta_{\min}(e))\Big)\Big\}.\end{aligned}$$

These bucketing estimators are finite-sample proxies intended to display refinement trends (they are not used in any formal proofs). Since the declared package V does not specify an uncertainty model or interval rule for $\widehat{\mathrm{Gap}}$, the plotted values are not licensed as numerical reports of $\mathrm{Gap}_D(\mathrm{End};\mathcal{C})$ for cross-run comparison.

6.4 Implementation note

A short Python script (no training) generates these plots by Monte Carlo sampling and endpoint bucketing. The script is intended as an optional appendix artifact; the theoretical statements in Sections 5 and 6 do not depend on it.

7 Results

This section reports purely *computational illustrations* of the proposed diagnostics. No model training is involved: the predictive distribution rule $h \mapsto p(\cdot \mid h)$ is defined explicitly by the controlled processes in Section 6.1, and gaps are estimated by endpoint bucketing as in Section 6.3.

All quantities plotted in this section are finite-sample proxies (denoted with hats, e.g., $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$). The declared threshold ε is shown only as a *reference level* for interpreting these proxies.In particular, observing $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C}) < \varepsilon$ in a plot is not, by itself, a package-licensed certificate that the true $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) < \varepsilon$.

7.1 Global fiber gaps versus refinement depth

We evaluate endpoint families parameterized by the number of additional scales/anchors K, and report the TV proxy $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ together with the auxiliary smoothed-entropy readout $\widehat{\mathrm{Gap}}_{\mathrm{KL,clip(EPS)},N}(\mathrm{End};\mathcal{C})$ from Section 6.3.

Process A (compressible dependence). For Process A, the predictive parameter $\theta_A(h)$ is controlled by a single slow statistic (a large-block parity). Endpoints that explicitly carry a compatible multi-scale statistic rapidly thin fibers, causing $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ to decrease with K and cross below the reference level ε in the proxy plot once the relevant scale is represented.By contrast, endpoints that add long-range information in coordinates misaligned with the controlling statistic can exhibit a persistent nonzero gap, illustrating that the TV proxy decreases only when the added endpoint coordinates track the operative long-range dependency.

Process B (blind-spot dependence). For Process B, the prediction depends on a single critical coordinate at lag $L^\star$. Endpoints that do not include that coordinate retain maximal within-fiber predictive variation, yielding $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C}) \sim 1$ (and a correspondingly large KL gap after clipping). Once the endpoint family includes an anchor that covers $L^\star$, the proxy plot for $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ crosses below the reference level ε (often sharply).This demonstrates a prototypical "blind-spot plateau": the proxy can exhibit a plateau and not decrease until a missing coordinate is explicitly carried.

7.2 Endpoint partition growth (refinement sanity check)

A structural sanity check is to track how refinement expands the induced endpoint partition. As K increases, the number of distinct endpoint values (buckets) observed under sampling increases in our simulations, consistent with refinement thinning fibers by splitting coarse equivalence classes.

Takeaway. Across both processes, the illustrations separate two regimes by a purely structural criterion: (i) aligned refinement, where added endpoint coordinates track the operative dependency and drive the TV proxy $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ into the sub-ε region, and (ii) misaligned refinement, where a missing critical coordinate yields a blind-spot plateau until the endpoint family explicitly carries the relevant information.These patterns are consistent with the refinement monotonicity principle (Proposition 1) together with the necessity obstruction in Proposition 3.

8 Discussion

This paper framed hierarchical and multi-scale mechanisms as *endpoint families* under a declared package V, and quantified within-endpoint ambiguity via fiber gaps $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$ and $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})$. We summarize the structural interpretation, what such mechanisms can and cannot guarantee, and practical implications for analysis.

8.1 Fiber thinning and predictive non-constancy

An endpoint map $\mathrm{End} : \mathcal{H} \to \mathcal{E}$ induces fibers $\mathrm{Fib}(e) = \mathrm{End}^{-1}(e)$, i.e., equivalence classes of histories that are indistinguishable at the endpoint level. The global gaps $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$ and $\mathrm{Gap}_{\mathrm{KL}}(\mathrm{End};\mathcal{C})$ quantify worst-case *predictive non-constancy within a fiber*: even if two histories share the same endpoint, their induced predictive distributions may differ substantially. In this

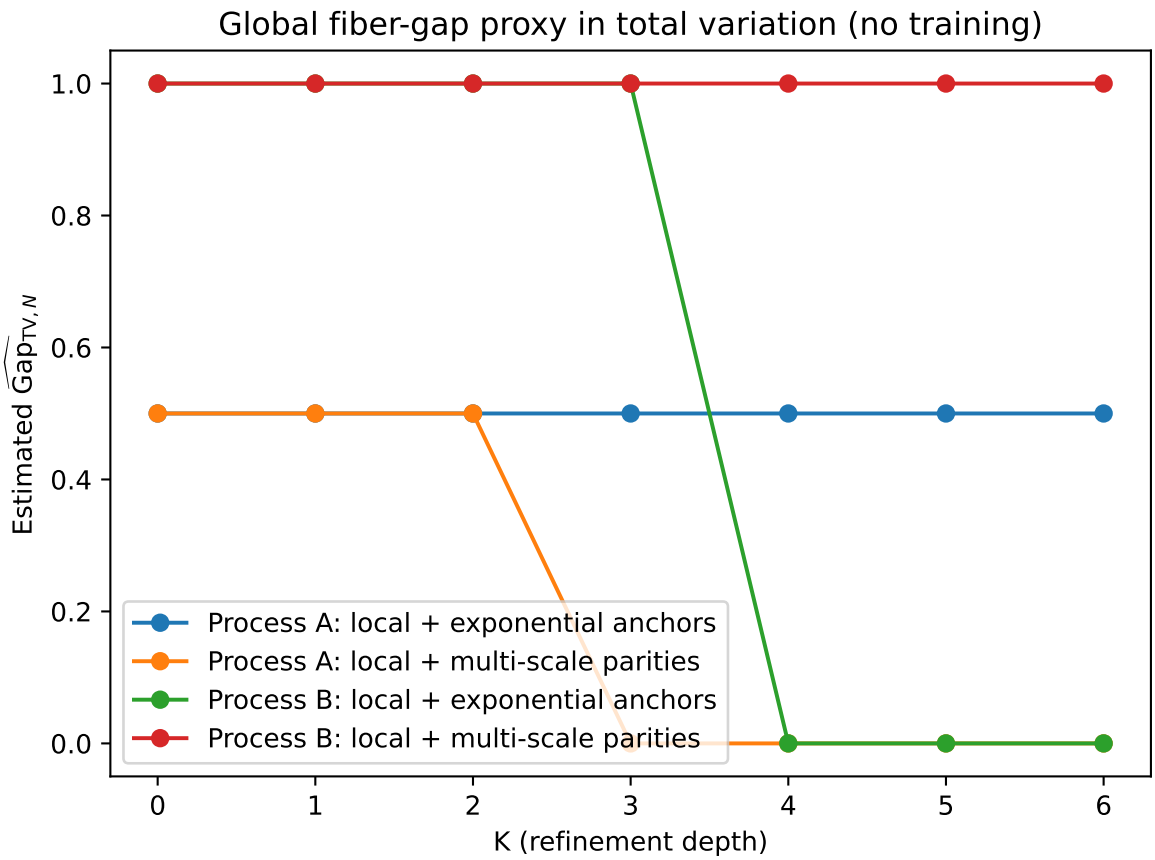

Figure 1: Global fiber-gap proxy in total variation, $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$, as a function of the refinement depth K, for the two controlled processes in Section 6.1 and the endpoint families in Section 6.2. The curves illustrate monotone refinement trends and blind-spot plateaus under the fixed declared package V. Here $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ is a finite-sample proxy used to display refinement trends. ε is shown only as the declared reference level; this plot does not certify ε-sufficiency and is not used as an input to any bound or theorem.

view, hierarchical mechanisms are ε-sufficient under the chosen V precisely when the *true* TV gap satisfies $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) < \varepsilon$, i.e., when the reporting convention $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}) \approx_\varepsilon 0$ applies on $\mathcal{C}$.

When hierarchical structure is implemented by concatenating additional deterministic coordinates to the endpoint (multi-scale summaries, anchor readings, memory tiers), the resulting endpoint family $\{\mathrm{End}_K\}$ refines the induced partition of $\mathcal{H}$. By Proposition 1, refinement implies monotone reduction of TV gaps and (when the KL readout is licensed under V) monotone reduction of KL gaps. The computational illustrations in Section 7 illustrate two qualitatively distinct regimes (at the proxy level): (i) aligned refinement, where added coordinates track the operative dependency and quickly suppress within-fiber variation, and (ii) misaligned refinement, where missing coordinates produce plateaus until the endpoint family explicitly carries the relevant information.

8.2 What hierarchical mechanisms buy

The diagnostic suggests a precise structural benefit: hierarchical mechanisms provide a controlled way to refine endpoints under finite budgets. Rather than attempting to preserve all history at full

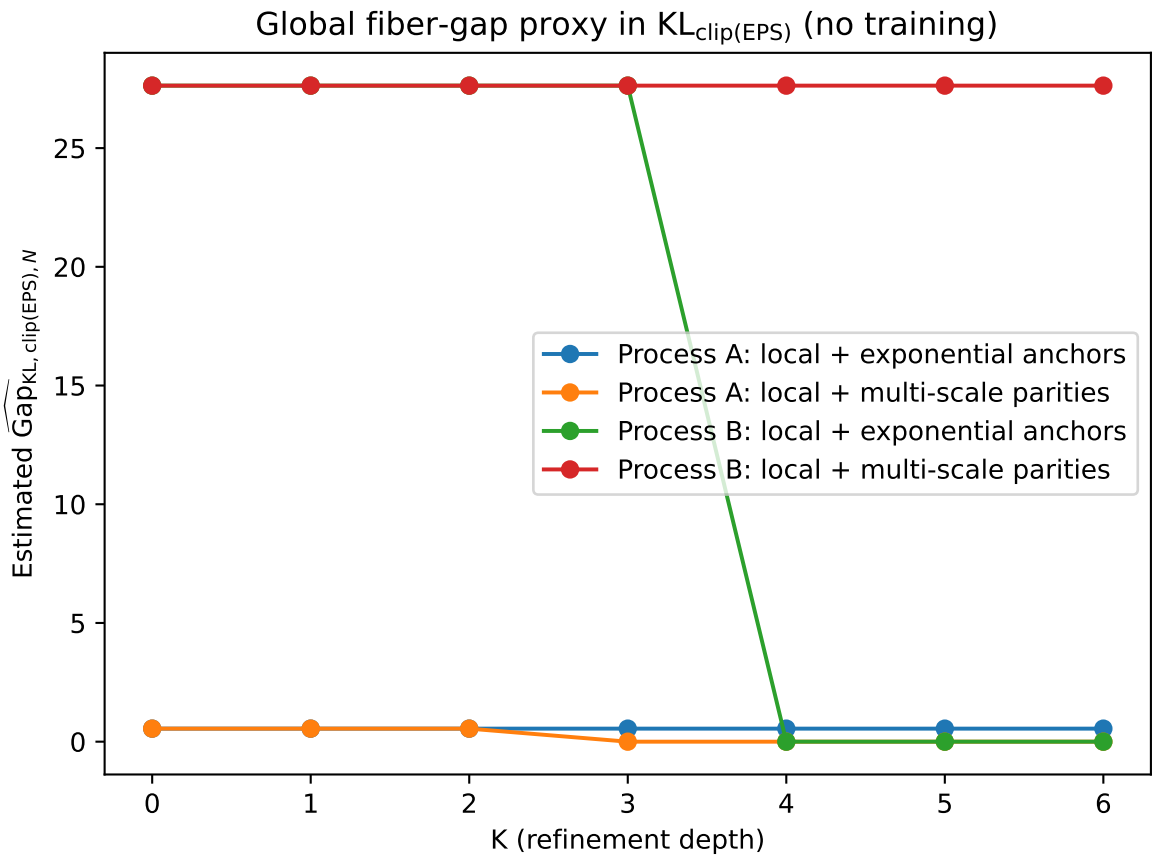

Figure 2: Auxiliary global fiber-gap proxy in smoothed relative entropy, $\widehat{\text{Gap}}_{\text{KL,clip(EPS)},N}(\text{End};\mathcal{C})$, as a function of refinement depth K. The readout uses $\text{KL}_{\text{clip(EPS)}}$ from Section 6.3 and is reported for visualization only; it is not used to derive TV-scale bounds for the original distributions.

resolution, one allocates representational capacity to a set of coarse but discriminative coordinates, which thins fibers and reduces predictive ambiguity under the chosen protocol V. This viewpoint is compatible with compression-based perspectives such as the information bottleneck [11]. This perspective also clarifies the role of multi-scale sparse designs: exponentially spaced anchors are not intrinsically superior, but offer an economical strategy for covering multiple distance scales with a small number of endpoint coordinates.

The KL-to-TV bridge (Proposition 2) provides a convenient separation of roles: KL often yields cleaner reasoning (e.g., compatibility with log-loss or information-theoretic manipulations), while TV offers bounded and interpretable effect sizes for reporting. Accordingly, when Gap_{KL} is licensed under V, one can reason in KL and present conclusions in TV without inflating notation or proof length. Relative entropy is also canonical in information geometry and renormalization/coarse-graining analyses [1, 15, 16].

8.3 What hierarchical mechanisms cannot buy: necessity and plateaus

Monotone improvement does not imply that, for arbitrarily small $\varepsilon > 0$, the protocol can achieve $\text{Gap}_{\text{TV}}(\text{End};\mathcal{C}) \approx_\varepsilon 0$.A minimal obstruction is endpoint capacity: Proposition 3 shows that if the endpoint space is too small relative to the intrinsic TV-separated diversity of the predictive family

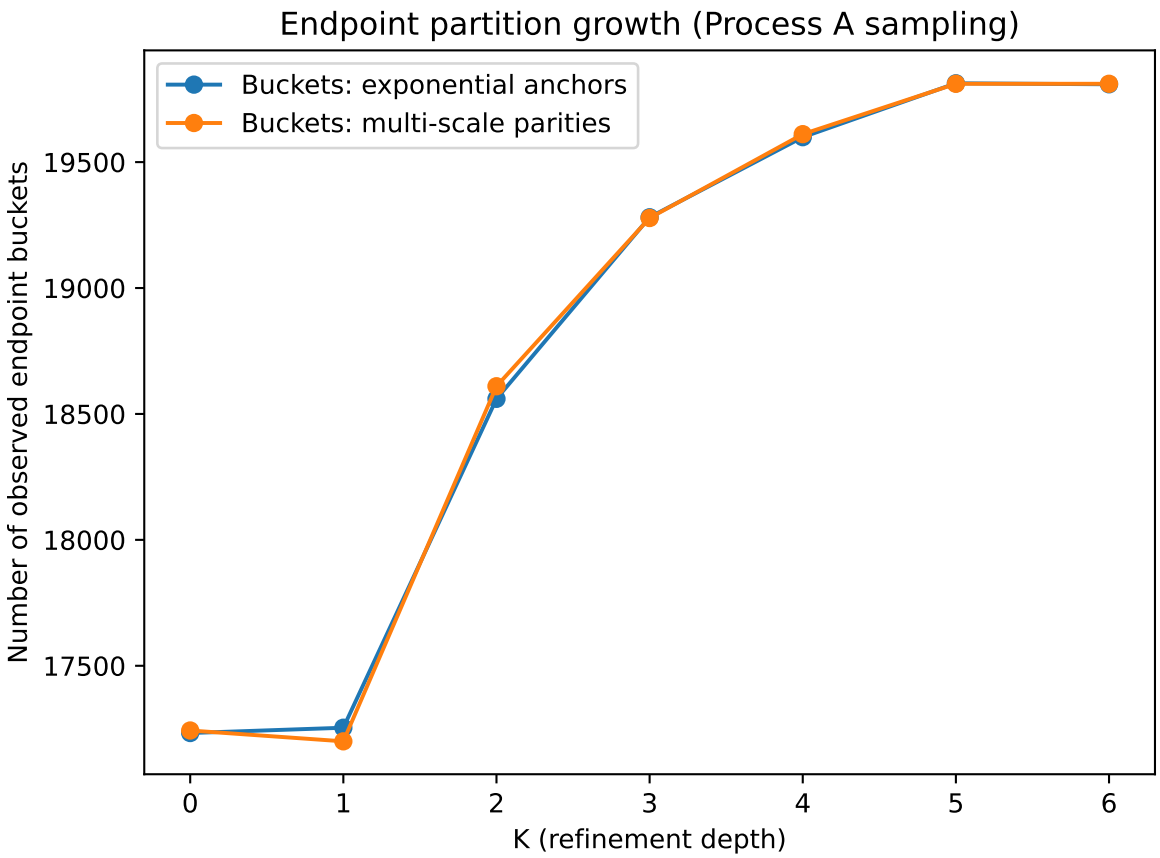

Figure 3: Observed number of endpoint buckets as a function of refinement depth K under Process A sampling (illustrative). Increasing bucket count indicates a finer partition of history space, consistent with endpoint refinement.

$\{p(\cdot \mid h)\}$, then $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}; \mathcal{C})$ must exceed a fixed lower bound. This yields a simple structural interpretation of plateaus: if the operative dependency cannot be represented by the available endpoint coordinates (either due to insufficient cardinality or due to structural blind spots), then within-fiber predictive variation persists under the fixed protocol V.

The blind-spot example in Section 7 illustrates a particularly sharp obstruction pattern: if prediction depends on a specific unobserved coordinate, then no amount of refinement that avoids that coordinate will reduce the gap. In more general settings, such obstructions correspond to missing features, missing scales, or compression operators that discard task-relevant degrees of freedom.

8.4 Implications for analysis and design

The proposed diagnostic is intended as an *analysis tool* rather than an architectural prescription. It suggests a practical workflow: fix V, choose a candidate endpoint family $\{\mathrm{End}_K\}$, and measure (or bound) the decay of $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End}_K; \mathcal{C})$. Rapid decay indicates that the added coordinates are aligned with the relevant dependencies under V, while plateaus indicate either insufficient endpoint capacity or structural misalignment. In this sense, endpoint design can be viewed as selecting which coordinates (summaries, anchors, tiers) to carry so that fibers become thin in the directions that

matter for the chosen declared package.

9 Conclusion

We introduced a declared-package diagnostic framework for understanding when hierarchical and multi-scale mechanisms can be ε-sufficient under a declared package V.By modeling such mechanisms as endpoint families $\{\mathrm{End}_K\}$ and quantifying within-endpoint ambiguity via fiber gaps, the framework makes ε-sufficiency statements well-posed under a fixed declared package $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$.A refinement monotonicity principle (Proposition 1) formalizes the structural effect of adding scales as fiber thinning, while a KL-to-TV bridge (Proposition 2) enables interpretable TV-scale reporting. A simple capacity lower bound (Proposition 3) explains why gap plateaus can persist when the endpoint space is too small or when refinement misses task-relevant coordinates.

The computational illustrations (no training) illustrate two regimes: aligned refinement can drive the TV proxy $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}(\mathrm{End};\mathcal{C})$ into the sub-ε region, whereas misaligned refinement exhibits blind-spot plateaus.Taken together, these results support a concise structural thesis: hierarchical mechanisms work insofar as they refine endpoints in directions that reduce within-fiber predictive non-constancy under the chosen protocol V, and they remain above-threshold when necessary degrees of freedom remain outside the endpoint representation.

Future work may extend the diagnostic to richer query families $\mathcal{Q}$, alternative discrepancy functionals, and empirical evaluations on trained models where $p(\cdot \mid h)$ is induced by learned representations rather than by explicit toy processes.

A Notation alignment and scope boundary (non-required)

This paper is self-contained. The material in this appendix is *not required* to read the main text, but records a compact scope boundary and a minimal notation-alignment with related diagnostics in [20, 19].

A.1 Fixed-slice scope (within-endpoint fibers)

All quantitative statements in the main text are *fixed-slice* (within-endpoint) diagnostics: they compare $p(\cdot \mid h)$ only among histories $h \in \mathcal{C}$ that share the same endpoint value $e = \mathrm{End}(h)$. The resulting objects are fiber diameters and global gaps, such as $\mathrm{Gap}_{\mathrm{TV}}(\mathrm{End};\mathcal{C})$, and the associated minimax residual $\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}, U_{\mathcal{E}})$.

A.2 Reserved symbols and non-identifications

The bare symbols Δ_I and Γ_{path} are reserved for endpoint–path (scale-chain) diagnostics in [19] and are not used here. This paper uses only within-endpoint (fixed-slice) objects—fiber diameters/gaps and the fixed-slice residual $\mathrm{Res}_{\mathrm{TV}}(\mathrm{End};\mathcal{C}, U_{\mathcal{E}})$—and does not identify any KL readout with endpoint–path gaps or Route-I divergences. Any bridge from fixed-slice fiber diagnostics to endpoint–path objects would require explicit path-process structure and a chain-rule decomposition along a scale chain, which are outside the scope of the present paper.

A.3 Minimal alignment map

The following correspondences are intended only as a naming alignment (not as a dependency of the main text):

- **Declared package V.** The tuple $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ is the local "declared protocol" used in this paper to gate domains, select reportable readouts, and fix the reporting threshold.
- **Fixed-slice residual $\mathrm{Res}_{\mathrm{TV}}$.** The symbol $\mathrm{Res}_{\mathrm{TV}}$ is used here for the fixed-slice minimax reconstruction residual $\mathrm{Res}_{\mathrm{TV}}(\mathrm{End}; \mathcal{C}, U_{\mathcal{E}})$ induced by a fixed endpoint map $\mathrm{End} : \mathcal{H} \to \mathcal{E}$ under the admissible reconstruction class $U_{\mathcal{E}}$ declared (via U) in $\mathcal{I}$.
- **Licensing discipline.** "Licensed under V" means the declared readout is finite on the domain-gated pairs relevant to the diagnostic (e.g. mutual absolute continuity for KL on fibers); otherwise the readout is treated as undefined under V.
- **Surrogate readouts.** Clipping-based proxies such as $\mathrm{KL}_{\mathrm{clip(EPS)}}$ are treated as separate declared numerical readouts for visualization and do not enter theorem statements about the original distributions.

B Optional Python Illustrations (Lightweight)

This appendix describes how to reproduce the figures in Section 7 using a short Python script (no training).

Exporting figures. In the provided script, replace `plt.show()` with the following lines to export files compatible with the main text:

```
plt.figure(1); plt.savefig("fig_gap_tv.pdf", bbox_inches="tight")
plt.figure(2); plt.savefig("fig_gap_kl.pdf", bbox_inches="tight")
plt.figure(3); plt.savefig("fig_endpoint_buckets.pdf", bbox_inches="tight")
```

Alternatively, export PNGs by replacing the `.pdf` suffix with `.png`.

File placement. Place the exported files (`fig_gap_tv.pdf`, `fig_gap_kl.pdf`, `fig_endpoint_buckets.pdf`) in the same directory as the LaTeX source, or update the paths in the corresponding `\includegraphics` commands.

B.1 Reference scripts

TV gap figure script.

```
"""fig_gap_tv.py

Estimated TV fiber-gap proxy (bucket-extrema) for Process A and Process B.

Index convention (matches the revised paper):
  A history is x = (x_0, ..., x_{T-1}). Python negative indices follow
  x[-d] == x_{T-d}.

Outputs:
  - fig_gap_tv.pdf
  - fig_gap_tv.png
"""

import numpy as np
import matplotlib.pyplot as plt
```

```python


# Sentinel for missing anchors (corresponds to \bot in the paper).
# Histories take values in {0,1}, so BOT=-1 is outside the token alphabet.
BOT = -1


# ---------------------------- Distances ----------------------------

def tv_bernoulli(p: float, q: float) -> float:
    """Total variation distance between Bernoulli(p) and Bernoulli(q)."""
    return abs(p - q)


# ---------------------------- Toy Processes -------------------------

def process_A_next_p(
    x: np.ndarray,
    big_block: int = 64,
    base: float = 0.25,
    alpha: float = 0.5,
) -> float:
    """Process A: p(x_{T} = 1 | history) in {0.25, 0.75} via a parity feature (amplitude
    parameter alpha)."""
    m = min(len(x), big_block)
    z = int(np.sum(x[-m:]) % 2)
    return base + alpha * z


def process_B_next_p(x: np.ndarray, lag_star: int = 128) -> float:
    """Process B: p(x_{T}=1|history) is (deterministic) x_{T-lag_star}."""
    if len(x) >= lag_star:
        return float(x[-lag_star])
    return 0.0


# ---------------------------- Endpoints ----------------------------

def end_local_window(x: np.ndarray, w: int):
    w_eff = min(len(x), w)
    return tuple(x[-w_eff:].tolist())


def end_local_plus_exp_anchors(x: np.ndarray, w: int, K: int):
    e = [end_local_window(x, w)]
    for j in range(K):
        d = w * (2**j)
        e.append(int(x[-d]) if len(x) >= d else BOT)
    return tuple(e)


def end_local_plus_multiscale_parities(x: np.ndarray, w: int, K: int):
    e = [end_local_window(x, w)]
    for j in range(K):
```

```python
        s = w * (2**j)
        m = min(len(x), s)
        parity = int(np.sum(x[-m:]) % 2)
        e.append(parity)
    return tuple(e)


# ---------------------------- Gap Estimation ------------------------

def estimate_gap_tv_hat(process_next_p, endpoint_fn, T: int, n_hist: int, seed: int) -> float:
    """\hat{Gap}_{TV,N}: bucket histories by endpoint and take max |p_max - p_min|."""
    rng = np.random.default_rng(seed)
    X = rng.integers(0, 2, size=(n_hist, T), dtype=np.int8)

    stats = {}  # endpoint -> [min_p, max_p]
    for i in range(n_hist):
        x = X[i]
        e = endpoint_fn(x)
        p = float(process_next_p(x))
        if e not in stats:
            stats[e] = [p, p]
        else:
            if p < stats[e][0]:
                stats[e][0] = p
            if p > stats[e][1]:
                stats[e][1] = p

    gap_tv = 0.0
    for pmin, pmax in stats.values():
        gap_tv = max(gap_tv, tv_bernoulli(pmin, pmax))
    return gap_tv


# ---------------------------- Main ---------------------------------

def main():
    # Parameters (kept consistent across scripts)
    T = 256
    n_hist = 20000
    w = 16
    Ks = list(range(0, 7))

    # Process A curves
    A_tv_exp, A_tv_par = [], []
    for K in Ks:
        A_tv_exp.append(
            estimate_gap_tv_hat(
                process_next_p=lambda x: process_A_next_p(x, big_block=64, base=0.25, alpha=0.5),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_exp_anchors(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=10 + K,
```

```
            )
        )
        A_tv_par.append(
            estimate_gap_tv_hat(
                process_next_p=lambda x: process_A_next_p(x, big_block=64, base=0.25, alpha=0.5),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_multiscale_parities(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=110 + K,
            )
        )

    # Process B curves
    lag_star = 128
    B_tv_exp, B_tv_par = [], []
    for K in Ks:
        B_tv_exp.append(
            estimate_gap_tv_hat(
                process_next_p=lambda x, lag_star=lag_star: process_B_next_p(x, lag_star=lag_star),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_exp_anchors(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=210 + K,
            )
        )
        B_tv_par.append(
            estimate_gap_tv_hat(
                process_next_p=lambda x, lag_star=lag_star: process_B_next_p(x, lag_star=lag_star),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_multiscale_parities(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=310 + K,
            )
        )

    # Plot
    fig = plt.figure()
    plt.plot(Ks, A_tv_exp, marker="o", label="Process A: local + exponential anchors")
    plt.plot(Ks, A_tv_par, marker="o", label="Process A: local + multi-scale parities")
    plt.plot(Ks, B_tv_exp, marker="o", label="Process B: local + exponential anchors")
    plt.plot(Ks, B_tv_par, marker="o", label="Process B: local + multi-scale parities")
    plt.xlabel("K (refinement depth)")
    plt.ylabel(r"Estimated $\widehat{\mathrm{Gap}}_{\mathrm{TV},N}$")
    plt.title("Global fiber-gap proxy in total variation (no training)")
    plt.legend()
    plt.tight_layout()

    # Save (names match LaTeX)
    fig.savefig("fig_gap_tv.pdf", bbox_inches="tight")
```

```
    fig.savefig("fig_gap_tv.png", dpi=300, bbox_inches="tight")
    plt.close(fig)


if __name__ == "__main__":
    main()
```

$\mathrm{KL}_{\mathrm{clip(EPS)}}$ gap figure script.

```
"""fig_gap_kl.py

Estimated KL_clip(EPS) fiber-gap proxy (bucket-extrema) for Process A and Process B.

Important:
  - Process B yields Bernoulli probabilities in {0,1}, so strict KL can be infinite.
  - This script reports a *declared* smoothed readout KL_clip(EPS) via clipping.

Index convention (matches the revised paper):
  A history is x = (x_0, ..., x_{T-1}). Python negative indices follow
  x[-d] == x_{T-d}.

Outputs:
  - fig_gap_kl.pdf
  - fig_gap_kl.png
"""

import numpy as np
import matplotlib.pyplot as plt


# Sentinel for missing anchors (corresponds to \bot in the paper).
# Histories take values in {0,1}, so BOT=-1 is outside the token alphabet.
BOT = -1


EPS = 1e-12


# ---------------------------- Distances ----------------------------

def kl_eps_bernoulli(p: float, q: float) -> float:
    """KL_clip(EPS)(Bern(p) || Bern(q)) with EPS-clipping for finite readout."""
    p = float(np.clip(p, EPS, 1 - EPS))
    q = float(np.clip(q, EPS, 1 - EPS))
    return p * np.log(p / q) + (1 - p) * np.log((1 - p) / (1 - q))


def sym_kl_eps_bernoulli(p: float, q: float) -> float:
    """Symmetrized KL_clip(EPS) used by the bucket-extrema proxy:
    max(KL_clip(EPS)(p||q), KL_clip(EPS)(q||p))."""
    return max(kl_eps_bernoulli(p, q), kl_eps_bernoulli(q, p))


# ---------------------------- Toy Processes ----------------------------
```

```python

def process_A_next_p(
    x: np.ndarray,
    big_block: int = 64,
    base: float = 0.25,
    alpha: float = 0.5,
) -> float:
    """Process A: p(x_{T} = 1 | history) in {0.25, 0.75} via a parity feature (amplitude parameter alpha)."""
    m = min(len(x), big_block)
    z = int(np.sum(x[-m:]) % 2)
    return base + alpha * z


def process_B_next_p(x: np.ndarray, lag_star: int = 128) -> float:
    """Process B: p(x_{T}=1|history) is (deterministic) x_{T-lag_star}."""
    if len(x) >= lag_star:
        return float(x[-lag_star])
    return 0.0


# ---------------------------- Endpoints ----------------------------

def end_local_window(x: np.ndarray, w: int):
    w_eff = min(len(x), w)
    return tuple(x[-w_eff:].tolist())


def end_local_plus_exp_anchors(x: np.ndarray, w: int, K: int):
    e = [end_local_window(x, w)]
    for j in range(K):
        d = w * (2**j)
        e.append(int(x[-d]) if len(x) >= d else BOT)
    return tuple(e)


def end_local_plus_multiscale_parities(x: np.ndarray, w: int, K: int):
    e = [end_local_window(x, w)]
    for j in range(K):
        s = w * (2**j)
        m = min(len(x), s)
        parity = int(np.sum(x[-m:]) % 2)
        e.append(parity)
    return tuple(e)


# ---------------------------- Gap Estimation ------------------------

def estimate_gap_kl_eps_hat_binary(process_label_fn, endpoint_fn, T: int, n_hist: int, seed: int, kl_const: float) -> float:
    """\hat{Gap}_{KL,EPS,N} specialized to binary-valued processes.

    In this toy setup, each process produces only two possible Bernoulli parameters.
    Therefore, the bucket-extrema KL_clip(EPS) proxy is either 0 (pure buckets) or
```

```python
    kl_const (a mixed bucket exists). We detect mixing via a 2-bit mask per bucket.
    """
    rng = np.random.default_rng(seed)
    X = rng.integers(0, 2, size=(n_hist, T), dtype=np.int8)

    seen = {}  # endpoint -> bitmask over labels {0,1}
    for i in range(n_hist):
        x = X[i]
        e = endpoint_fn(x)
        z = int(process_label_fn(x))  # z in {0,1}
        seen[e] = seen.get(e, 0) | (1 << z)

    return kl_const if any(mask == 3 for mask in seen.values()) else 0.0


# ---------------------------- Main ----------------------------------

def main():
    # Parameters (kept consistent across scripts)
    T = 256
    n_hist = 20000
    w = 16
    Ks = list(range(0, 7))

    # Constants for mapping labels -> Bernoulli parameters
    # Process A: p in {0.25, 0.75}
    pA0, pA1 = 0.25, 0.75
    KL_A = sym_kl_eps_bernoulli(pA0, pA1)
    # Process B: p in {0, 1} (after smoothing for KL_clip(EPS))
    pB0, pB1 = 0.0, 1.0
    KL_B = sym_kl_eps_bernoulli(pB0, pB1)

    # Process A curves (binary label = parity feature)
    A_kl_exp, A_kl_par = [], []
    for K in Ks:
        A_kl_exp.append(
            estimate_gap_kl_eps_hat_binary(
                process_label_fn=lambda x: int(np.sum(x[-min(len(x), 64):]) % 2),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_exp_anchors(x, w, K),
                T=T, n_hist=n_hist, seed=10 + K,
                kl_const=KL_A,
            )
        )
        A_kl_par.append(
            estimate_gap_kl_eps_hat_binary(
                process_label_fn=lambda x: int(np.sum(x[-min(len(x), 64):]) % 2),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_multiscale_parities(x, w, K),
                T=T, n_hist=n_hist, seed=110 + K,
                kl_const=KL_A,
            )
        )

    # Process B curves (binary label = anchor bit at lag_star, with default 0)
```

```python
    lag_star = 128
    B_kl_exp, B_kl_par = [], []
    for K in Ks:
        B_kl_exp.append(
            estimate_gap_kl_eps_hat_binary(
                process_label_fn=lambda x, lag_star=lag_star: (int(x[-lag_star]) if len(x) >= lag_star else 0),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_exp_anchors(x, w, K),
                T=T, n_hist=n_hist, seed=210 + K,
                kl_const=KL_B,
            )
        )
        B_kl_par.append(
            estimate_gap_kl_eps_hat_binary(
                process_label_fn=lambda x, lag_star=lag_star: (int(x[-lag_star]) if len(x) >= lag_star else 0),
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_multiscale_parities(x, w, K),
                T=T, n_hist=n_hist, seed=310 + K,
                kl_const=KL_B,
            )
        )

    # Plot
    fig = plt.figure()
    plt.plot(Ks, A_kl_exp, marker="o", label="Process A: local + exponential anchors")
    plt.plot(Ks, A_kl_par, marker="o", label="Process A: local + multi-scale parities")
    plt.plot(Ks, B_kl_exp, marker="o", label="Process B: local + exponential anchors")
    plt.plot(Ks, B_kl_par, marker="o", label="Process B: local + multi-scale parities")
    plt.xlabel("K (refinement depth)")
    plt.ylabel(r"Estimated $\widehat{\mathrm{Gap}}_{\mathrm{KL,clip(EPS)},N}$")
    plt.title("Global fiber-gap proxy in $\mathrm{KL}_{\mathrm{clip(EPS)}}$ (no training)")
    plt.legend()
    plt.tight_layout()

    # Save (names match LaTeX)
    fig.savefig("fig_gap_kl.pdf", bbox_inches="tight")
    fig.savefig("fig_gap_kl.png", dpi=300, bbox_inches="tight")
    plt.close(fig)


if __name__ == "__main__":
    main()
```

Endpoint bucket figure script.

```python
"""fig_endpoint_buckets.py

Endpoint bucket-count diagnostic (Process A sampling).

Index convention (matches the revised paper):
  A history is x = (x_0, ..., x_{T-1}). Python negative indices follow
  x[-d] == x_{T-d}.
```

```python

Outputs:
  - fig_endpoint_buckets.pdf
  - fig_endpoint_buckets.png
"""

import numpy as np
import matplotlib.pyplot as plt


# Sentinel for missing anchors (corresponds to \bot in the paper).
# Histories take values in {0,1}, so BOT=-1 is outside the token alphabet.
BOT = -1


# ---------------------------- Endpoints ----------------------------

def end_local_window(x: np.ndarray, w: int):
    """Local window endpoint: (x_{T-w}, ..., x_{T-1}) with truncation."""
    w_eff = min(len(x), w)
    return tuple(x[-w_eff:].tolist())


def end_local_plus_exp_anchors(x: np.ndarray, w: int, K: int):
    """Local window + exponentially spaced anchors x_{T-w*2^j}, j=0..K-1."""
    e = [end_local_window(x, w)]
    for j in range(K):
        d = w * (2**j)
        e.append(int(x[-d]) if len(x) >= d else BOT)
    return tuple(e)


def end_local_plus_multiscale_parities(x: np.ndarray, w: int, K: int):
    """Local window + parities of suffix blocks of size w*2^j, j=0..K-1."""
    e = [end_local_window(x, w)]
    for j in range(K):
        s = w * (2**j)
        m = min(len(x), s)
        parity = int(np.sum(x[-m:]) % 2)
        e.append(parity)
    return tuple(e)


# ---------------------------- Bucket Counting ------------------------

def count_buckets(endpoint_fn, T: int, n_hist: int, seed: int) -> int:
    rng = np.random.default_rng(seed)
    X = rng.integers(0, 2, size=(n_hist, T), dtype=np.int8)
    buckets = set()
    for i in range(n_hist):
        buckets.add(endpoint_fn(X[i]))
    return len(buckets)


```

```python
# ---------------------------- Main ---------------------------------

def main():
    # Parameters (kept consistent across scripts)
    T = 256
    n_hist = 20000
    w = 16
    Ks = list(range(0, 7))  # refinement depth

    bins_exp, bins_par = [], []
    for K in Ks:
        bins_exp.append(
            count_buckets(
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_exp_anchors(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=500 + K,
            )
        )
        bins_par.append(
            count_buckets(
                endpoint_fn=lambda x, w=w, K=K: end_local_plus_multiscale_parities(x, w, K),
                T=T,
                n_hist=n_hist,
                seed=600 + K,
            )
        )

    fig = plt.figure()
    plt.plot(Ks, bins_exp, marker="o", label="Buckets: exponential anchors")
    plt.plot(Ks, bins_par, marker="o", label="Buckets: multi-scale parities")
    plt.xlabel("K (refinement depth)")
    plt.ylabel("Number of observed endpoint buckets")
    plt.title("Endpoint partition growth (Process A sampling)")
    plt.legend()
    plt.tight_layout()

    # Save (names match LaTeX)
    fig.savefig("fig_endpoint_buckets.pdf", bbox_inches="tight")
    fig.savefig("fig_endpoint_buckets.png", dpi=300, bbox_inches="tight")
    plt.close(fig)


if __name__ == "__main__":
    main()
```

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